CS6660: Mathematical Foundations of Data Science (Probability)

Practice Problems 01

Indian Institute of Technology Hyderabad

Topics: Sample Space, σ-Algebras, Probability Measures and Their Properties

1. Let $\mathscr{F}_1$ and $\mathscr{F}_2$ be two σ-algebras on a common sample space Ω.
   (a) Is $\mathscr{F}_1 \cap \mathscr{F}_2$ necessarily a σ-algebra? If yes, prove formally. If not, provide a counterexample.
   (b) Is $\mathscr{F}_1 \cup \mathscr{F}_2$ necessarily a σ-algebra? If yes, prove formally. If not, provide a counterexample.

2. Consider a probability space $(\Omega, \mathscr{F}, \mathbb{P})$.
   Let $A_1, A_2, \ldots \in \mathscr{F}$ satisfy the property that $\mathbb{P}(A_k) = 1$ for all $k \in \mathbb{N}$. Show that
   $$\mathbb{P}\left(\bigcap_{k\in\mathbb{N}} A_k\right) = 1.$$
   As a corollary, show that if $A, B \in \mathscr{F}$ are such that $\mathbb{P}(A) = 1$ and $\mathbb{P}(B) = 1$, then $\mathbb{P}(A \cap B) = 1$.

3. Out of all the students in a class, 60% wear glasses, 70% watch Sherlock, and 40% belong to both the categories. Determine the probability that a student selected uniformly at random neither wears glasses nor watches Sherlock.

4. Let $\Omega = \{1, \ldots, p\}$ where p is a prime number. Let $\mathscr{F} = 2^{\Omega}$, and let $\mathbb{P}$ be defined on $(\Omega, \mathscr{F})$ as
   $$\mathbb{P}(A) = \frac{|A|}{p}, \qquad A \in \mathscr{F},$$
   where $|A|$ denotes the number of elements in A.
   Derive the necessary and sufficient conditions for two sets A and B to be independent.

5. Let Ω be a sample space, and let $\mathscr{F}$ be a σ-algebra of subsets of Ω. Fix $B \in \mathscr{F}$, and consider the collection
   $$\mathscr{G} = \{A \cap B : A \in \mathscr{F}\}.$$
   That is, $\mathscr{G}$ is a collection of subsets of B formed by taking the intersection of each set in $\mathscr{F}$ with B.
   Show that $\mathscr{G}$ is a σ-algebra of subsets of B.

6. Fix $n \in \mathbb{N}$. Suppose that an experiment with sample space Ω is performed repeatedly n times.
   For any set $E \in \mathscr{F}$, let $n(E)$ denote the number of times that event E occurs in the n trials of the experiment.
   Let $f : \mathscr{F} \to [0, 1]$ be defined as
   $$f(E) = \frac{n(E)}{n}, \quad E \in \mathscr{F}.$$
   Show that f satisfies the axioms of probability, and is therefore a valid probability measure on $(\Omega, \mathscr{F})$.

7. Suppose two fair coins are tossed independently of each other.
   (a) Specify $(\Omega, \mathscr{F}, \mathbb{P})$ for the above experiment.
   (b) Find the probability of the event that both coins turn up heads, conditioned on the event that the first coin turns up head.
   (c) Find the probability of the event that both coins turn up heads, conditioned on the event that at least one of the coins turns up head.

8. Consider events $A, B, C \in \mathscr{F}$ such A is independent of B and A is independent of C.
   Show that A is independent of $B \cup C$ if and only if A is independent of $B \cap C$.
   Note: To prove an if and only if statement, the "if" and "only if" directions must be proved separately.